

Report
Of
LLM Post-Training: Data Preparation & Evaluation

Submitted
To
**School of Computer Science and Engineering IILM
University, Greater Noida, U.P.**

In partial fulfilment of the requirement of degree of

B.Tech CSE



By
Roll Number of Student: CS-2341654
Name of Student: Vaibhav Narayan
Batch: 2026-27

CANDIDATE'S DECLARATION

I, Vaibhav Narayan, do hereby declare that the internship report titled “LLM Post Training: Data Preparation & Evaluation” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: Vaibhav Narayan
Roll No.: CS-2341654

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to specially thank Ms. Usha Pandey, Head of Human Resources at Ethara AI, for her guidance and support throughout the internship, and the faculty of the Department of Computer Science and Engineering at IILM University, Greater Noida, for facilitating this internship as part of the curriculum.

I express my gratitude to my parents for their blessings.

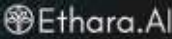
Signature

Student Name: Vaibhav Narayan
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3. INTERNSHIP COMPLETION CERTIFICATE

 | Supercharge your LLM with human aligned data

Internship Completion Letter

Name: Vaibhav Narayan
Email: vaibhavnarayan@gmail.com

Dear Vaibhav Narayan,

Congratulations on successfully completing your internship with Ethara AI. We appreciate the dedication and professionalism you demonstrated throughout your tenure.

Your internship was conducted from **06/02/2026** to **12/05/2026**. During this period, you contributed towards data preparation and evaluation tasks in the AI and data projects supporting Large Language Model (LLM) post-training workflows. We are confident that the experience gained during this program will support your future professional growth.

We wish you every success in your future endeavours. Please feel free to contact us at hr@ethara.ai if you require any further assistance.

Sincerely,
Ethara AI
Usha Pandey



Head, Human Resources
hr@ethara.ai

Acknowledgement and Acceptance

I, **Vaibhav Narayan** accept the Internship Completion Letter for the position of LLM Post Training Intern as outlined in this letter.

Signature:

Date: 15th June 2026

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4. PROJECT DESCRIPTION

4.1 Introduction

This report summarizes the work undertaken during my internship at Ethara AI, carried out from February 6, 2026 to May 12, 2026 as part of the academic requirements of my B.Tech program in Computer Science and Engineering at IILM University, Greater Noida. The internship offered exposure to the practical pipeline behind training and refining Large Language Models (LLMs), with a focus on data preparation and human evaluation — two of the most critical, yet often underappreciated, stages of LLM posttraining. Alongside these tasks, I also worked on foundational AI engineering activities and learned, hands-on, how AI systems are actually developed and built.

4.2 Organization Profile

Ethara AI works on data and evaluation pipelines that support the post-training of Large Language Models. Post-training — the stage after a model's initial pretraining — relies heavily on carefully curated datasets and structured human feedback to align model behavior with desired standards of quality, helpfulness, and safety. Ethara AI's work sits at this intersection of data engineering and human-in-the-loop evaluation.

4.3 Problem Statement

Raw data collected for training language models is frequently noisy, duplicated, inconsistently formatted, or of variable quality, while model outputs generated during post-training must be reliably judged against one another to guide alignment. Without disciplined data cleaning and consistent, criteria-based response evaluation, posttraining efforts risk reinforcing the very flaws they are meant to correct. This internship addressed both problems in a practical setting.

4.4 Project Objectives

- To understand the end-to-end data pipeline that feeds into LLM post-training.
- To gain hands-on experience in cleaning and preparing raw data for model training.
- To learn structured methods for evaluating and comparing model-generated responses.

- To build a working understanding of how large language models function internally.
- To explore AI engineering as a career discipline and identify the skills it demands.

4.5 Scope of the Project

The scope of the internship was limited to two functional areas of the LLM post-training pipeline: (a) cleaning and preparing text/conversational datasets destined for model training, and (b) performing comparative, criteria-based evaluation of paired model responses to identical prompts. Model architecture design, training infrastructure, and deployment were outside the scope of this internship.

4.6 Technologies and Tools Used

- Large Language Model (LLM) outputs and conversational datasets as the primary data medium.
- Structured spreadsheets/annotation interfaces for data cleaning and labelling.
- Comparative (pairwise) evaluation rubrics based on correctness, relevance, clarity, and helpfulness — the human-feedback methodology used in RLHF style alignment.

4.7 System Architecture

At a conceptual level, the workflow followed a linear pipeline: Raw Data Collection → Data Cleaning & Filtering → Curated Training Dataset → Model Post-Training (Supervised Fine-Tuning and Preference Alignment) → Paired Response Generation → Human Comparative Evaluation → Feedback Signal fed back into Model Alignment. My contributions sat at the Data Cleaning and Human Comparative Evaluation stages of this pipeline.

4.8 Methodology

- **Data Cleaning:** Reviewed raw text and conversational data, removed duplicate and malformed entries, corrected formatting inconsistencies, and flagged content that did not meet the required quality bar.
- **Comparative Evaluation:** Reviewed pairs of model-generated responses to the same prompt and judged which response was better using a consistent rubric covering correctness, relevance, clarity, and helpfulness.
- **AI Engineering Practice:** Undertook foundational AI engineering tasks alongside the core data and evaluation work, gaining direct, hands-on exposure to how AI systems are actually developed — from preparing data and setting up evaluation workflows to observing how these pieces come together to improve a model.
- **Conceptual Study:** Studied how LLMs are trained in two broad stages — pretraining on large text corpora, followed by post-training (supervised finetuning and preference-based alignment) — and how human feedback loops shape model behavior.

4.9 AI Engineering Fundamentals Learned

Beyond the two core work streams above, the internship also gave me a working, hands-on introduction to the broader discipline of AI engineering. The key fundamentals I learned and practiced include:

- **Data Engineering:** Collecting, cleaning, and labelling raw text/conversational data so that it is fit for model training.
- **Model Basics:** How LLMs are structured internally, and the practical difference between pretraining and post-training.
- **Evaluation & Feedback:** Building consistent evaluation rubrics and comparing model outputs — the RLHF-style human-feedback loop used to align model behavior.
- **Prompting & Tooling:** Working with prompts, annotation platforms, spreadsheets, and other everyday AI development tools.
- **Debugging & Iteration:** Spotting poor-quality outputs, tracing them back to underlying data or prompt issues, and refining the work accordingly.

- Documentation & Collaboration: Reporting findings clearly and coordinating with the team and HR throughout the internship.

4.10 Expected Outcomes

- A working understanding of how data quality directly determines LLM output quality.
- Practical experience applying a consistent rubric to judge and compare AI-generated responses.
- Hands-on exposure to AI engineering practices and a clearer, practical understanding of how AI systems are developed end-to-end.
- A conceptual grasp of the LLM training lifecycle, from pretraining through post-training alignment.
- Clarity on AI engineering as a broad discipline spanning data engineering, evaluation design, and model alignment — informing my career direction toward AI/ML engineering roles.

4.11 Certificates of Completion and Other Communication Proof

The Internship Completion Certificate issued by Ethara AI is attached in Chapter 3 of this report. All internship-related communication was conducted via official email and HR channels with Ms. Usha Pandey, Head of Human Resources at Ethara AI.

5. BIBLIOGRAPHY/REFERENCES

1. Ethara AI — Internship Completion Certificate and internal task communication, 2026.
2. Ouyang, L. et al., “Training language models to follow instructions with human feedback,” Advances in Neural Information Processing Systems (NeurIPS), 2022.
3. Bai, Y. et al., “Training a Helpful and Harmless Assistant with Reinforcement Learning from Human Feedback,” arXiv:2204.05862, 2022.